Math 571 - Homework 4

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Problem 4.1 (B:4.2:2)**.** Show that the following are equivalent for $f : X \to Y$, where X and Y are metric spaces.

(a) f is continuous.

(b) If (p_n) is a sequence from X and $p_n \to p$, then $f(p_n) \to f(p)$, equivalently, $\lim_n f(p_n) = f(\lim_n p_n)$.

(c) If (p_n) is convergent, then $f(p_n)$ is convergent.

Hint: For (c) $\implies$ (b) let $q = \lim_n f(p_n)$ and consider $(p_0, p, p_1, p, p_3, p, \dots) = (\hat{p}_n)$, then $(f(\hat{p}_n)) = (f(p_0), f(p), f(p_1), f(p), \dots)$ must converge.

(a) $\implies$ (b) Suppose f is continuous (at p) If $p_i \to p$, then for any $\epsilon > 0$ find $\delta > 0$ so that $f(N_\delta(p)) \subseteq N_\epsilon(f(p))$. Let N be so that for all $i > N$, $d(p_i, p) < \delta$, then $f(p_i) \in N_\epsilon(f(p))$ for all $i > N$. So $f(p_i) \to f(p)$.

(b) $\implies$ (c) This is trivial.

(c) $\implies$ (b) Assume (c). Let $p_i \to p$ consider $(p_1, p, p_2, p, \dots)$, then clearly $p_i \to p$ and by assumption $(f(p_1), f(p), f(p_2), f(p), \dots)$ converges, but this must converge to $f(p)$. So (b) holds.

(b) $\implies$ (a) Let $\epsilon > 0$. Suppose f is not continuous at p, then there is $\delta_1 > \delta_2 > \dots > 0$ so that $\delta_i \to 0$ and there is $p_i \in N_{\delta_i}(p)$ with $p_i \notin f(N_\epsilon(p))$. But this would mean $p_i \to p$ and $f(p_i) \not\to f(p)$.

Problem 4.2 (B:4.2:3)**.** Let X be a metric space.

(a) Show that there are open subsets A, B of X, such that $A \cup B = X$ and $A \cap B = \emptyset$ if and only if there exists a continuous function $f : X \to R$ such that $f(a) = 0$ for all $a \in A$ and $f(b) = 1$ for all $b \in B$.

(b) Deduce that X is connected if and only if every continuous function $X \to \{0, 1\}$ is constant.

(c) Show likewise that X is connected if and only if every continuous function $X \to \mathbb{Z}$ is constant.

(a) If such open sets A and B exist, then these are actually clopen and hence closed sets. Problem 5.9 below provides a canonical $f : X \to [0, 1]$ so that $f(A) = \{0\}$ and $f(B) = \{1\}$. Conversely, if we had such an f, then $f^{-1}(\{0\}) = A$ and $f^{-1}(\{1\}) = B$ give our sets.

(b) and (c) and more. If X is connected and $f : X \to Y$ is continuous, then $f(X)$ is connected. If Y is discrete, then the only way $f(X)$ can be connected is if $f(X) = \{y\}$ so f is constant.

Problem 4.3 (B:4.2:4)**.** (a) Suppose $f : \mathbb{R} \to \mathbb{R}$ is a continuous function such that $f(x) > x$ for all x. For any positive integer n, let f^n denote the n-fold composite $f \circ f \circ \ldots \circ f$. Show that for all $x, y \in R$, there exists a positive integer n such that $f^n(x) > y$.

(b) Does the conclusion of part (a) remain true if the assumption that f is continuous is removed?

(a) Suppose $x < y$, if $f^n(x) \leq y$ for all n, then $(f^n(x))$ is an increasing sequence bounded by y and so $\lim_{n\to\infty} f^n(x) = x^* \leq y$ exists. Since f is continuous, and hence sequentially continuous,

$$f(x^*) = f\big(\lim_{n\to\infty} f^n(x)\big) = \lim_{n\to\infty} f(f^n(x)) = \lim_{n\to\infty} f^{n+1}(x) = x^*$$

This contradicts the assumption that $f(x^*) > x^*$.

(b) We can define a function where some $f^n(x)$ is bounded and $f(x) > x$ for all x. One way would be

$$f(x) = \begin{cases} \frac{p+1}{q+1} & \text{if } x = \frac{p}{q} \text{ for } 0 < p < q \text{ integers} \\ x+1 & \text{otherwise} \end{cases}$$

It is easy to check that for $0 < p < q$, $\frac{p}{q} < \frac{p+1}{q+1}$ and that $0 < p+1 < q+1$. So $f^n\big(\frac{p}{q}\big) = \frac{p+n}{q+n} \to 1$. It is also clear that $f(x) > x$ for all x. So letting $x = \frac{1}{2}$ and $y = 1$ we have $f^n\big(\frac{1}{2}\big) < 1$. This shows that (a) does not hold for this f.

Problem 4.4 (R:4.11)**.** Suppose f is a uniformly continuous mapping of a metric space X into a metric space Y and prove that $(f(x_n))$ is a Cauchy sequence in Y for every Cauchy sequence (x_n) in X. Use this result to give an alternative proof of the theorem stated in Exercise 13.

Fix $\epsilon > 0$ and $\delta > 0$ so that for all $x, y \in X$, $d_X(x,y) < \delta \implies d_Y(f(x), f(y)) < \epsilon$. For all $i, jk > M$, $d_X(x_i, x_j) < \delta$ and hence for all $i, j > M$, $d_Y(f(x_i), f(x_j)) < \epsilon$. So we see that $(f(x_i))$ is a Cauchy sequence.

Note that if f is not uniformly continuous, for example, $f(x) = \frac{1}{x}$ on $(0, \infty)$. Then $(\frac{1}{n}) \mapsto (n)$ while $(n) \mapsto (\frac{1}{n})$, so Cauchy sequences and non-Cauchy sequences can have Cauchy images.

Problem 4.5 (R:4.13)**.** Let E be a dense subset of a metric space X, and let f be a uniformly continuous *real* function defined on E. Prove that f has a continuous extension from E to $\mathrm{cl}(E)$. (This means, there is a unique continuous $\hat{f} : \mathrm{cl}(E) \to Y$ so that $\hat{f} \supset f$.)

Could the range space $\mathbb{R}^1$ be replaced by $\mathbb{R}^k$? By any compact metric space? By any complete metric space? By any metric space?

Better Hint: Use R:4.11

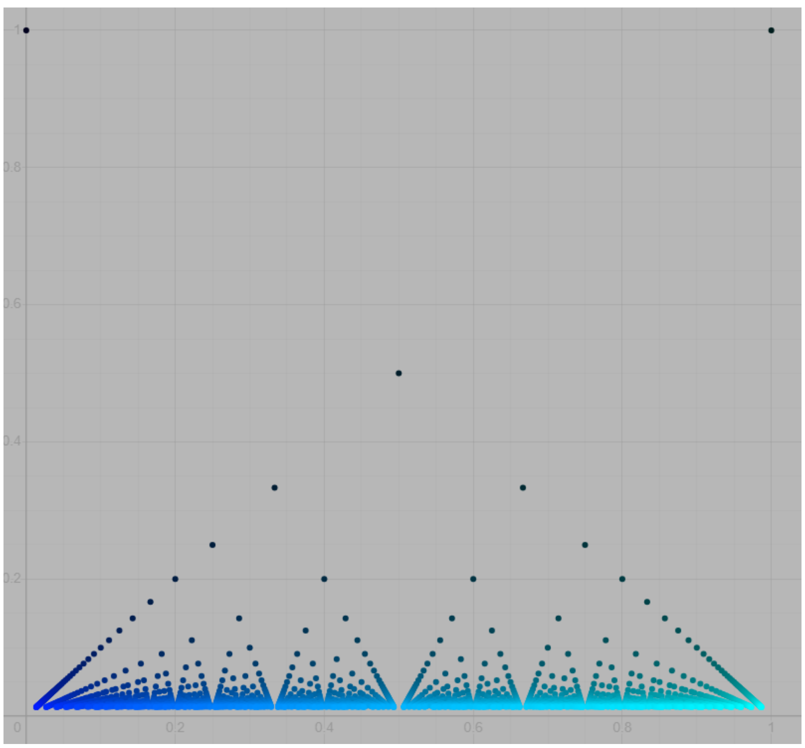
We can look at any $f : E \to Y$ where Y is complete. Take $x \in \mathrm{cl}(E)$ and (x_i) a sequence from E with $x = \lim_i x_i$. Then $(f(x_i))$ is a Cauchy sequence and as Y is complete $f(x_i) \to y$. We try defining $f(x) = y$ in this case. We need to see that this is well defined, that is, if $x_i \to x$ and $x'_i \to x$, $f(x_i) \to y$, and $f(x'_i) \to y'$, then $y = y'$.

This is clear as the *merged sequence* (z_i) where $z_{2i} = x_i$ and $z_{2i+1} = x'_i$ is also a Cauchy sequence and clearly $y = \lim_i f(x_i) = \lim_i f(z_i)$ and $y' = \lim_i f(x'_i) = \lim_i f(z_i)$, so $y = y'$.

So we have defined $\hat{f} : \mathrm{cl}(E) \to Y$ that extends f. Now take $\epsilon > 0$ and take δ so that $d_X(x, y) < \delta \implies d_Y(f(x), f(y)) < \epsilon$. We want to see that this holds with $\hat{f}$ in place of f. So suppose $x, y \in \mathrm{cl}(E)$ and $d(x, y) < \delta$. Then if $x_i, y_i \in E$ with $x_i \to x$ and $y_i \to y$. We can find M so that for $i > M$, $d_X(x_i, y_i) < \delta$ and hence $d_Y(f(x_i), f(y_i)) < \epsilon$, but then $d_Y(\hat{f}(x), \hat{f}(y)) = d_Y(\lim_i f(x_i), \lim_i f(y_i)) = \lim_i d_Y(f(x_i), f(y_i)) < \epsilon$. So $\hat{f} : \mathrm{cl}(E) \to Y$ is actually uniformly continuous.

Problem 4.6 (R:4.18)**.** Define the ***Thomae Function*** as follows

$$f(x) = \begin{cases} 0 & x \text{ is irrational} \\ \frac{1}{n} & x = \frac{m}{n}, \text{ where } n, m \in \mathbb{Z}, n > 0, \text{ and } \gcd(m, n) = 1 \end{cases}$$



Desmos

(a) Show that f is discontinuous at exactly the rational points.

(b) Show that for $g : \mathbb{R} \to \mathbb{R}$, $\mathrm{Dis}(f) = \{x \mid f \text{ is discontinuous at } x\}$ is an F_σ set. (A union of countably many closed sets.)

Hint: Look at B:4.3:7 and the ***oscillation*** of g at x, denoted $\omega_f(x)$. Show that $\mathrm{Dis}(f) =$

$\bigcup D_n$ where $D_n = \{x \mid \omega_g(x) \geq \frac{1}{n}\}$ and that D_n is closed.

(c) Show that there is no $g : \mathbb{R} \to \mathbb{R}$ that is continuous at exactly the irrational points. (Baire Category)

(a) If $r = \frac{m}{n}$ with $\gcd(m, n) = 1$ and $n > 0$, then $f(r) = \frac{1}{n}$ but we can find (α_i) irrational so that $\alpha_i \to r$, but then $f(\alpha_i) \to 0 \neq f(r) = \frac{1}{n}$. So f is discontinuous at each rational.

To prove the converse, it is good to notice that $f(x) = f(x+1)$, for irrational numbers, this is clear. For $x = \frac{m}{n}$ we have $x + 1 = \frac{m+n}{n}$ and $\gcd(m+n, n) = \gcd(m, n) = 1$. So f is ***periodic*** with period 1.

Conversely, let $\alpha \in (0, 1)$ be irrational. It suffices to show that f is continuous at α by periodicity. For any $\epsilon > 0$, there is $s \in \mathbb{N}$ so that $\frac{1}{s} < \epsilon$. For $i = 1, 2, \ldots, s$ we find k_i so that $\frac{k_i}{i} < \alpha < \frac{k_i+1}{i}$ let $\delta = \inf_i \min\{|\alpha - \frac{k_i}{i}|, |\alpha - \frac{k_i+1}{i}|\}$. For $|\alpha - \frac{m}{n}| < \delta$ we have $\frac{k_i}{s} < \frac{m}{n} < \frac{k_i+1}{s}$ and thus $n > s$ and so $f\left(\frac{m}{n}\right) = \frac{1}{n} < \epsilon$. So $|\alpha - \beta| < \delta \implies |f(\alpha) - f(\beta)| < \epsilon$.

(b) To see that $\mathrm{Dis}(f) = \{x \mid x \text{ is a discontinuity of } f\}$ is an F_σ requires a bit of work. This is related to B:4.3.7. Define the ***oscillation*** of f at x by $\omega_f(x) = \lim_{\delta \to 0^+} \mathrm{diam}(f(N_\delta(x))$ since for $\delta_1 > \delta_2 > 0$ we have $\mathrm{diam}(f(N_{\delta_2}(x)) \leq \mathrm{diam}(f(N_{\delta_1}(x))$ it is clear this limit always exists. This measure *how discontinuous f is at x*. Clearly,

$$f \text{ is continuous at } x \iff \omega_f(x) = 0$$

So $D = \{x \mid \omega_f(x) > 0\} = \bigcup_n D_n$ where $D_n = \{x \mid \omega_f(x) \geq \frac{1}{n}\}$. Once we see that D_n is closed, then it is clear that D is F_σ.

To see that D_n is closed, let $x_n \in D_n$ with $x_n \to x$. We must see that $\omega_f(x) \geq \frac{1}{n}$. We know for $i > N$, $x_i \in N_\delta(x)$ and so there is $\delta' < \delta$ so that $N_{\delta'}(x_i) \subseteq N_\delta(x)$, but then $\mathrm{diam}(f(N_\delta(x)) \geq \mathrm{diam}(N_{\delta'}(x_i)) \geq \frac{1}{n}$. Since δ was arbitrary, $\omega_f(x) \geq \frac{1}{n}$ and so $x \in D_n$. Thus D_n is closed.

(c) For any function the set of discontinuities is an F_σ set. It is true that every F_σ subset of a metric space can be realized as the set of discontinuities of $f : X \to \mathbb{R}$. Since $\mathbb{R} \setminus \mathbb{Q}$ is not an F_σ by the Baire Category Theorem, there is no function $f : \mathbb{R} \to \mathbb{R}$ whose set of discontinuities is exactly the irrationals.

Problem 4.7 (B:4.4:3)**.** Amazingly, there are continuous functions $f : [0, 1] \to [0, 1]^k$ for any k. Here are a couple of nice videos on this topic:

- A great 3Blue1Brown video: *Hilbert's Curve: Is infinite math useful?*
- Eddie Woo discussing several constructions in a high school class: *Space-Filling Curves (1 of 4: Peano Curve)*

(a) Let $n > 1$. Show that for every point p of $[0, 1]^n$, the subset obtained by removing p from $[0, 1]^n$ is still connected. On the other hand, point to a result in Rudin showing that $[0, 1]$ does not have this property.

(b) Show how the existence of a one-to-one continuous map from $[0, 1]$ onto $[0, 1]^n$, together with the above pair of facts, would lead to a contradiction. (Hint: What do you know about one-to-one continuous maps of one compact metric space onto another?)

Here we are given $f : [0,1] \to [0,1]^k$ that is continuous and surjective.

(a) Suppose $[0,1]^k \setminus \{p\}$ is clearly path-connected for $k > 0$ and hence connected.

Path-Connected $\implies$ Connected Suppose X is path-connected, that is, For all $p, q \in X$ there is continuous $f : [0,1] \to X$ with $f(0) = p$ and $f(1) = q$. If X is not connected, then get disjoint non-empty open sets A and B with $A \cap B = \emptyset$. Let $a \in A$ and $b \in B$, take $f : [0,1] \to X$ continuous with $f(0) = 1$ and $f(1) = b$, then $f^{-1}(A) = A_0$ and $f^{-1}(B) = B_0$ are non-empty disjoint open subsets on $[0,1]$ with $[0,1] = A_0 \cup B_0$ so $[0,1]$ would be disconnected, a contradiction.

(b) If f is bijective, then as $[0,1]$ is compact, it is an open map and so $f^{-1} : [0,1]^k \to [0,1]$ is continuous. But then $f([0,1]^k \setminus \{p\}) = [0,1] \setminus \{f(p)\}$ is connected. But this would mean that $f(p) \in \{0,1\}$, **for all** p!

Problem 4.8 (B:4.3:4)**.** (a) Suppose K is a compact metric space, and $f : K \to K$ is a map with the property that for every pair of distinct points $x, y \in K$ one has $d(f(x), f(y)) < d(x,y)$. Show that there exists a unique point $p \in K$ such that $f(p) = p$.

(b) Show by examples that the result of (a) is false if we put any of the noncompact metric spaces $[0,1)$, $[0,+\infty)$ or $\mathbb{R}$ in place of the compact space K.

(a) First notice that f is continuous, since $d(x,y) < \epsilon \implies d(f(x), f(y)) < \epsilon$.

Existence: Pick any initial point x_0, define $x_{i+1} = f(x_i)$. By compactness, there is a subsequence x_{n_i} converging to some p. We see $f(p) = f(\lim_i x_{n_i}) = \lim_i f(x_{n_i}) = \lim_i x_{n_{i+1}}$. Now $d(p, f(p)) = \lim_i d(x_{n_i}, x_{n_{i+1}})$ and if we look at $c_i = d(x_i, x_{i+1})$ we see that $c_i \geq c_{i+1}$ is a descending sequence and so has a limit c and so $d(p, f(p)) = c$. But we can apply f one more time to get $f(f(p)) = f(\lim_i x_{n_{i+1}}) = \lim_i f(x_{n_{i+1}}) = \lim_i x_{n_{i+2}}$ and as above $d(f(p), f(f(p))) = \lim_i d(x_{n_{i+1}}, x_{n_{i+1}}) = c$. So we have $d(p, f(p)) = d(f(p), f(f(p))) = c$, but by assumption, this means $p = f(p)$

Uniqueness: Suppose $f(p) = p$ and $f(q) = q$, then $d(p,q) = d(f(p), f(q))$, but as in the existence portion, this means $p = q$. So the fixed point is unique.

(b) Take $f(x) = \frac{1}{2}(1+x)$ so $f : [0,1) \to [\frac{1}{2}, 1)$. Clearly, $|x-y| > |f(x) - f(y)| = \frac{1}{2}|x-y|$ also clear is that $f(x) \neq x$ for any $x \in [0,1)$. The exact same idea can be used for the other two examples.

Problem 4.9 (R4.20/22)**.** Let X be a metric space, and for any nonempty subset $E \subset X$, define the distance from a point $x \in X$ to E as $\rho_E(x) = \inf_{z \in E} d(x,z)$.

(a) Prove that $\rho_E(x)$ is a uniformly continuous function on X, and that $\rho_E(x) = 0$ if and only if $x \in \overline{E}$.

(b) Let A and B be disjoint, nonempty closed sets in X. Define:

$$f(p) = \frac{\rho_A(p)}{\rho_A(p) + \rho_B(p)}$$

Prove that f is a continuous function on X with range in $[0,1]$, such that $f(p) = 0$ precisely on A and $f(p) = 1$ precisely on B.

(a) To show that $\rho_E(x)$ is uniformly continuous, let $\epsilon > 0$. Suppose $0 < d(y,x) < \epsilon/2$. Take $z \in E$ with $d(x,z) < \rho_E(x) + \epsilon/2$, then $\rho_E(y) \leq d(y,z) \leq d(y,x) + d(x,z) < \rho_E(x) + \epsilon$ and so $\rho_E(y) - \rho_E(x) < \epsilon$. Similarly, take $z' \in E$ with $\rho_E(y) < d(y,z') + \epsilon/2$, then $\rho_E(x) \leq d(x,z') \leq d(x,y) + d(y,z') < \rho_E(y) + \epsilon$. Thus we have $|\rho_E(x) - \rho_E(y)| < \epsilon$. We have shown that **for any** y,

$$d(x,y) < \epsilon/2 \implies |\rho_E(x) - \rho_E(y)| < \epsilon$$

Thus $\delta = \epsilon/2$ works ***uniformly*** for the $\epsilon - \delta$ definition of continuous and so $x \mapsto \rho_E(x)$ is a uniformly continuous function.

It is also clear that if $z \in \text{cl}(E)$ and $z_i \in E$ are chosen so that $z_i \to z$, then $\rho_E(z) = 0$. Conversely, if $\rho_E(z) = 0$, choose $z_i \in E$ so that $d(z, z_i) \to 0$, then $z_i \to z$ and so $z \in \text{cl}(E)$.

(b) From part (a) we know that if $p \in A$, then $f(x) = \frac{0}{0+\rho_B(p)} = 0$ where $\rho_B(p) > 0$, since $p \notin \text{cl}(B) = B$. Similarly, for $p \in B$ we have $f(p) = \frac{\rho_A(p)}{\rho_A(p)+0} = 1$ since $\rho_A(p) \neq 0$. Also we know that if $p \notin A \cup B$, then $\rho_A(p) > 0$ and $\rho_B(p) > 0$ and so $0 < \frac{\rho_A(p)}{\rho_A(p)+\rho_B(p)} < 1$.

For continuity, fix p and $\epsilon > 0$ choose q so that

$$d(p,q) < \delta \implies |\rho_A(p) - \rho_A(q)| < \epsilon_1 \wedge |\rho_B(p) - \rho_B(q)| < \epsilon_1$$

where ϵ_1 is determined later. Then

$$\begin{aligned}
|f(p) - f(q)| &= \left| \frac{\rho_A(p)}{\rho_A(p) + \rho_B(p)} - \frac{\rho_A(q)}{\rho_A(q) + \rho_B(q)} \right| \\
&= \left| \frac{\rho_A(p)(\rho_A(q) + \rho_B(q)) - \rho_A(q)(\rho_A(p) + \rho_B(p))}{(\rho_A(p) + \rho_B(p))(\rho_A(q) + \rho_B(q))} \right| \\
&= \left| \frac{\rho_A(p)\rho_B(q) - \rho_A(q)\rho_B(p))}{(\rho_A(p) + \rho_B(p))(\rho_A(q) + \rho_B(q))} \right| \\
&= \left| \frac{\rho_A(p)\rho_B(q) - \rho_A(q)\rho_B(q) + \rho_A(q)\rho_B(q) - \rho_A(q)\rho_B(p))}{(\rho_A(p) + \rho_B(p))(\rho_A(q) + \rho_B(q))} \right| \\
&= \left| \frac{\rho_B(q)(\rho_A(p) - \rho_A(q)) + \rho_A(q)(\rho_B(q) - \rho_B(p))}{(\rho_A(p) + \rho_B(p))(\rho_A(q) + \rho_B(q))} \right| \\
&\leq \left(\frac{\rho_B(q)}{(\rho_A(p) + \rho_B(p))(\rho_A(q) + \rho_B(q))} \right) |\rho_A(p) - \rho_A(q)| \\
&\quad + \left(\frac{\rho_A(q)}{(\rho_A(p) + \rho_B(p))(\rho_A(q) + \rho_B(q))} \right) |\rho_B(q) - \rho_B(p)| \\
&\leq \left(\frac{\rho_B(q)}{(\rho_A(p) + \rho_B(p))(\rho_A(q) + \rho_B(q))} \right) \epsilon_1 + \left(\frac{\rho_A(q)}{(\rho_A(p) + \rho_B(p))(\rho_A(q) + \rho_B(q))} \right) \epsilon_1 \\
&\leq \left(\frac{\rho_A(q) + \rho_B(q)}{(\rho_A(p) + \rho_B(p))(\rho_A(q) + \rho_B(q))} \right) \epsilon_1 \\
&= \frac{\epsilon_1}{\rho_A(p) + \rho_B(p)}
\end{aligned}$$

Now we set ϵ_1. Let $\epsilon_1 < (\rho_A(p) + \rho_B(p))\epsilon$ then we have $|f(p) - f(q)| < \epsilon$.

Note: In the above we chose δ so that $|\rho_A(p) - \rho_A(q)| < \epsilon(\rho_A(p) + \rho_B(p))$ and $|\rho_B(p) - \rho_B(q)| < \epsilon(\rho_A(p) + \rho_B(p))$. This δ depends on p so we are not showing that f is uniformly continuous and there are examples where f is not uniformly continuous, e.g., take $X = \mathbb{R}$, $A = \{n \in \mathbb{N} \mid n \geq 2\}$, and $B = \{n + \frac{1}{n} \mid n \in \mathbb{N} \wedge n \geq 2\}$. So A and B are closed and disjoint. Take $x_n = n$ and $y_n = n + \frac{1}{n}$, then $d(x_n, y_n) \to 0$, but $f(x_i) = 1$ and $f(y_i) = 1$. If f were uniformly continuous, then for some $\delta > 0$, $d(x, y) < \delta \implies |f(x) - f(y)| < \frac{1}{2}$. But this clearly fails for the sequence defined.

You could skip this, but it adds context.

The next problem uses a tool that is also at the heart of some of the space-filling curves and is a good capstone to the techniques in this chapter. Consider the space $C([0,1]) = \{f : [0,1] \to \mathbb{R} \mid f \text{ is continuous}\}$. (You have studied this as an inner product space in MAT-512). X is provided with a norm, the L_∞***-norm*** or ***uniform*** norm, which is different from the L_2-norm, $\|f\|_2 = \left(\int_0^1 f^2\,dx\right)^{1/2}$ coming from the inner-product $\langle f, g\rangle = \int_0^1 fg\,dx$. The uniform norm is given by

$$\|f\|_1 = \sup\{|f(x)| \mid x \in [0,1]\}$$

This gives rise, as all norms do, to a metric, $d_1 : C([0,1]) \times C([0,1]) \to [0,\infty)$ given by $d_1(f,g) = \|f - g\|_1 = \sup\{|f(x) - g(x)| \mid x \in [0,1]\}$. It should be clear that the 1/2-norms and the resulting metrics are just the natural generalization to $C([0,1]) \subset \mathbb{R}^{[0,1]}$ of the corresponding norms/metrics on $\mathbb{R}^k$. (In fact, this is what you naturally do when approximating continuous functions by taking test points, a technique common to both Calculus and computer science.)

Note: By the extreme value theorem, $\sup\{|f(x)| \mid x \in [0,1]\}$ can be replaced by $\max\{|f(x)| \mid x \in [0,1]\}$.

Convergence in $C([0,1])$ under the L_∞-norm is called ***uniform convergence***, i.e., convergence in the uniform norm. Clearly, $f_i \to f$ in $C([0,1])$ iff for all $\epsilon > 0$, there is $N > 0$, so that for all $i > N$, $\sup\{|f_i(x) - f(x)| \mid x \in [0,1]\} < \epsilon$.

Fact: $C([0,1])$ under the 1-norm metric is complete.

Proof: It is actually an excellent exercise to prove this. Suppose (f_i) is a Cauchy sequence. Then $(f_i(x))$ is a Cauchy sequence for each x ans since $\mathbb{R}$ is complete, $f_i(x) \to y$ for each x, so consider $f(x) = \lim_i f_i(x)$. To complete the argument, one only needs to show that $f_i \to f$ in the 1-norm, i.e., uniformly. Fix $\epsilon > 0$ and N so that $\|f_i - f_m\|_1 < \epsilon/2$ for $i, m > N$. Then $|f_i(x) - f(x)| \leq |f_i(x) - f_m(x)| + |f_m(x) - f(x)|$. By choosing $m > N$ with $|f_m(x) - f(x)| < \epsilon/2$ we see that $|f_i(x) - f(x)| < \epsilon$ for all $i > N$ for all $x \in [0,1]$. But then $\|f_i - f\|_1 < \epsilon$ for all $i > N$ so $f_i \to_1 f$.

You should stop skipping if you chose to skip above.

We can rephrase uniform convergence as $f_i \to f$ uniformly if

$$(\forall \epsilon > 0)(\exists N > 0)(\forall i > N)(\forall x \in [0,1])(|f_i(x) - f(x)| < \epsilon) \quad (\dagger)$$

Note that $f_i(x) \to f(x)$ ***pointwise*** iff

$$(\forall \epsilon > 0)(\forall x \in [0,1])(\exists N > 0)(\forall i > N)(|f_i(x) - f(x)| < \epsilon)$$

So uniform convergence means we can find for each ϵ and N that ***works uniformly*** for all $x \in [0,1]$.

Main Result: If f_i are continuous and $f_i \to f$ uniformly in the sense of (†), then f is continuous.

Proof. This would have made a nice exercise in itself. For $\epsilon > 0$ and take $N > 0$ so that for all $i > N$, $|f_i(x) - f(x)| < \frac{\epsilon}{3}$ for all $x \in [0,1]$. Take $i > N$ and find δ so that $0 < |y - x| < \delta \implies |f_i(y) - f_i(x)| < \frac{\epsilon}{3}$. Now take $0 < |y - x| < \delta$, then

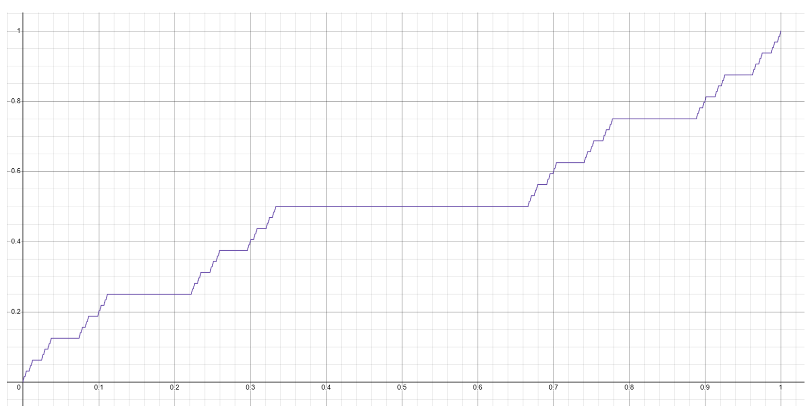
$$|f(y) - f(x)| \leq |f(y) - f_i(y)| + |f_i(y) - f_i(x)| + |f_i(x) - f(x)| < \epsilon$$

by the triangle inequality. So we have verified that

$$0 < |y - x| \implies |f(y) - f(x)| < \epsilon$$

Note: If $f_i \to f$ pointwise, but not uniformly, then f need not be continuous. Consider $f_n(x) = x^n$ on $[0,1]$, $f(x) = 0$ for $x \in [0,1)$, but $f(1) = 1$, so f is not continuous. This shows that $f_n \to f$ pointwise, but not uniformly.

Problem 4.10 (**The Cantor Function**)**.** Let $C \subseteq [0,1]$ be the standard Cantor ternary set. Let $f_n : [0,1] \to [0,1]$ be the sequence of continuous, monotonically increasing functions where $f_0(x) = x$, and each f_{n+1} is constructed by modifying f_n so that it is constant on the middle thirds removed in the $(n+1)$-th step of the Cantor set construction.



Desmos

(a) Prove that the sequence $\{f_n\}$ converges uniformly on $[0,1]$ to a limit function f (the Cantor function).

Hint: You may use that $C([0,1])$ is complete as a metric space using the uniform metric. This way it suffices to show $\|f_n - f_m\|_\infty < \epsilon$ for sufficiently large m, n.

(b) Prove that f is continuous and monotonically non-decreasing on $[0,1]$.

(c) Prove that f is surjective onto $[0,1]$, yet is locally constant on the complement of C (an open set of measure 1).

(a): Define $f_n(x)$ to be piecewise linear recursively:

- $f_0(x) = x$
- Given $f_n(x)$ define $f_{n+1}(x)$ by

$$f_n(x) = \begin{cases} \frac{1}{2} f_n(3x) & \text{if } 0 \le x \le \frac{1}{3} \\ \frac{1}{2} & \text{if } \frac{1}{3} < x < \frac{2}{3} \\ \frac{1}{2} + \frac{1}{2} f_n\big(3\big(x - \frac{2}{3}\big)\big) & \text{if } \frac{2}{3} \le x \le 1 \end{cases}$$

Clearly, each f_n is piecewise linear and monotonic with no jump discontinuities and hence is continuous. By construction, $\|f_n - f_m\| = 2^{-n}$ for $m > n$ so (f_n) is a Cauchy sequence under the 1-norm metric. It follows that $(f_n(x))$ is a Cauchy sequence for each $x \in [0,1]$ and by completeness, $f_n(x) \to y$ for some y for each x. Define $f(x) = \lim_n f_n(x)$, then $f : [0,1] \to [0,1]$.

(b): It is clear that $|f(x) - f_n(x)| \le |f(x) - f_m(x)| + |f_n(x) - f_m(x)| < 2^{-n} + 2^{-n} = 2^{-n+1}$ for sufficiently large n and so $\|f(x) - f_n(x)\|_1 \to 0$, that is, $f_n \to f$ uniformly. We know from the Main Fact above that $f(x)$ is continuous.

Since f is a limit of monotonic functions, it too is monotonic. For if $y < x$ and $f(x) < f(y)$ we could find large enough m so that $f_m(x) < f_m(y)$. This would be a contradiction. Here, we used only pointwise convergence.

(c): The Cantor function is constant on the open set consisting of the intervals removed. We know the lengths of these open intervals are $\frac{1}{3} + \frac{2}{3^2} + \frac{2^2}{3^3} + \cdots$, the sum of this series is $\frac{1}{3}\big(\frac{1}{1-2/3}\big) = 1$ so the open intervals on which f is constant has *length*, or ***measure***, 1.

Note: I basically re-proved the completeness fact mentioned above in (a); this was just to reinforce that argument. You could have verified (a) and (b) simultaneously by simply showing that (f_n) is a Cauchy sequence in the 1-norm metric and then applying the completeness proved above to see that $f_n \to f$ uniformly for some $f \in C([0,1])$.